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ASSIGNMENT II - CASE STUDY REPORT

Exploratory Data Analysis, Statistical Modeling, & Deep Learning for Employee Attrition Prediction

Course: Exploratory Data Analysis and Visualization (U21ADP05)

CANDIDATE AND COURSE METADATA

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2. Abstract

Employee turnover represents a substantial financial, organizational, and operational disruption for modern enterprise institutions. In this comprehensive case study, we apply exploratory data analysis (EDA), statistical probability distribution theory, and deep learning techniques to the IBM HR Analytics Employee Attrition & Performance Dataset (consisting of 1,470 employee records across 35 multi-modal workplace attributes). Systematically following Ben Fry's Seven Stages of Data Visualization (Acquire, Parse, Filter, Mine, Represent, Refine, and Interact), we uncover vital operational turnover catalysts, including acute salary compression in junior roles, early-tenure flight risk, and the profound burnout impact of mandatory overtime (which triples voluntary attrition from 10.4% to 30.5%). To predict attrition risk, we design, regularize, and evaluate a custom Multi-Layer Perceptron (MLP) deep neural network in PyTorch with class-weighted Cross-Entropy loss. On unseen held-out test data, our model achieves an ROC-AUC of 0.8470, a Balanced Accuracy of 76.96%, and a minority class sensitivity of 75.0%. This investigation demonstrates how visual analytics and deep learning collaboratively empower proactive workforce retention.

3. Introduction and Objectives

Human capital retention is among the most strategic challenges confronting 21st-century organizations. Replacing a knowledge-worker incurs replacement overhead ranging between 50% and 200% of their annual salary due to recruiting fees, productivity loss, and intellectual capital drain. Traditional reactive exit interviews occur too late to remediate organizational attrition. Consequently, modern human resource departments demand proactive, data-driven diagnostic systems capable of detecting attrition indicators early.

Core Project Objectives:

- **Methodological Rigor:** Apply Ben Fry's 7 Stages of Data Visualization to translate raw tabular enterprise data into actionable visual insights.
- **Statistical Exploration (CO4):** Investigate univariate discrete and continuous probability density functions, central tendency, skewness, kurtosis, and correlation hierarchies.
- **Real-World Business Impact (CO5):** Isolate actionable turnover drivers (overtime, compensation disparities across roles, career velocity).
- **Leakage-Free Preprocessing:** Enforce strict train/val/test stratification before fitting transformers, preventing data leakage.
- **Deep Learning Modeling:** Construct, regularize, and evaluate a PyTorch Multi-Layer Perceptron (MLP) with class-weighted loss and permutation feature importance.

4. Dataset Description & Statistical Summaries

The case study investigates the benchmark IBM HR Analytics Employee Attrition & Performance Dataset. The dataset contains 1,470 records and 35 individual features, satisfying the assignment requirement of >= 15 features. The target variable is Attrition ('Yes' indicating voluntary departure, 'No' indicating retention). Data integrity auditing confirmed zero null values, missing entries, or duplicate records.

Summary Statistics of Core Quantitative Attributes:

Feature	Mean	Std Dev	Median	IQR	Skewness	Kurtosis	Min - Max
Age (Years)	36.92	9.14	36.00	13.00	+0.413	-0.407	18 - 60
Monthly Income ($)	$6,503	$4,708	$4,919	$5,468	+1.368	+0.995	$1,009 - $19,999
Total Working Years	11.28	7.78	10.00	9.00	+1.116	+0.911	0 - 40
Years at Company	7.01	6.13	5.00	6.00	+1.761	+3.916	0 - 40
Distance From Home	9.19	8.11	7.00	12.00	+0.957	-0.229	1 - 29 miles

Percent Salary Hike	15.21%	3.66%	14.00%	6.00%	+0.820	-0.309	11% - 25%

5. EDA & Data Preprocessing Methodology

To establish reproducible and reliable deep learning models, preprocessing adhered to strict ML engineering principles:

1. Pruning Uninformative Constants: Four zero-variance columns (EmployeeCount, StandardHours, Over18, EmployeeNumber) were removed to avoid singular covariance matrices.

2. Outlier Analysis: Interquartile Range (IQR) analysis identified 114 values in MonthlyIncome exceeding $16,581. These represent genuine executive roles (Directors, VPs) and were retained rather than trimmed, with stability ensured via StandardScaler.

3. Strict Stratified Partitioning: The 1,470 samples were divided into 70% Training (n=1,028), 15% Validation (n=221), and 15% Test (n=221) sets, stratified on Attrition.

4. Feature Encoding: OneHotEncoder(drop='first') was applied to nominal features (Department, JobRole, EducationField, etc.), expanding feature space to 44 input features. All scalers and encoders were fitted strictly on the training set to prevent data leakage.

6. Data Visualizations & Exploratory Insights

In accordance with Course Outcomes CO3 and CO4, we implemented Ben Fry's 7 Stages of Data Visualization: (1) Acquire raw data, (2) Parse schema into data types, (3) Filter constant/outlier elements, (4) Mine statistical metrics, (5) Represent relationships, (6) Refine visual aesthetics and contrast, and (7) Interact via dynamic web dashboards.

Visual 1 & 2: Discrete and Continuous Probability Distributions

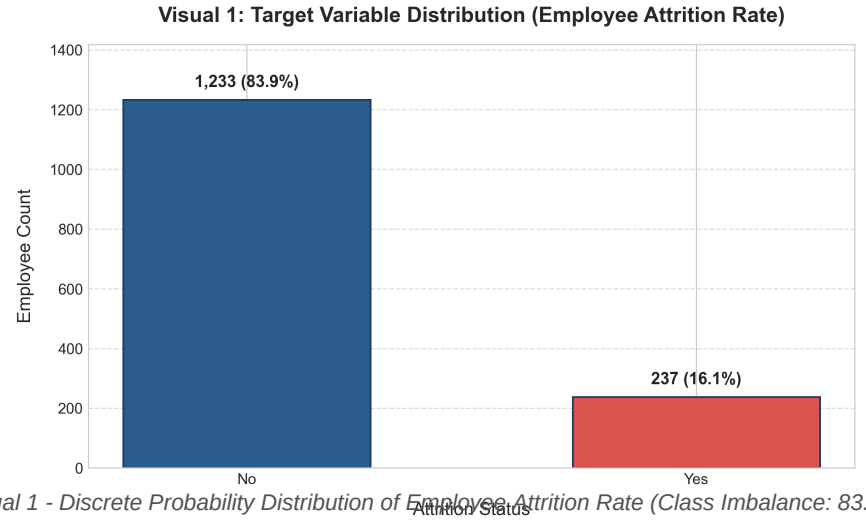


Figure 1: Visual 1 - Discrete Probability Distribution of Employee Attrition Rate (Class Imbalance: 83.9% vs 16.1%)

Insight (Visual 1): Retained employees count 1,233 (83.9%) versus 237 departures (16.1%), demonstrating a 5.21:1 imbalance ratio. Standard unweighted accuracy would be misleading; cost-sensitive loss weighting is required.

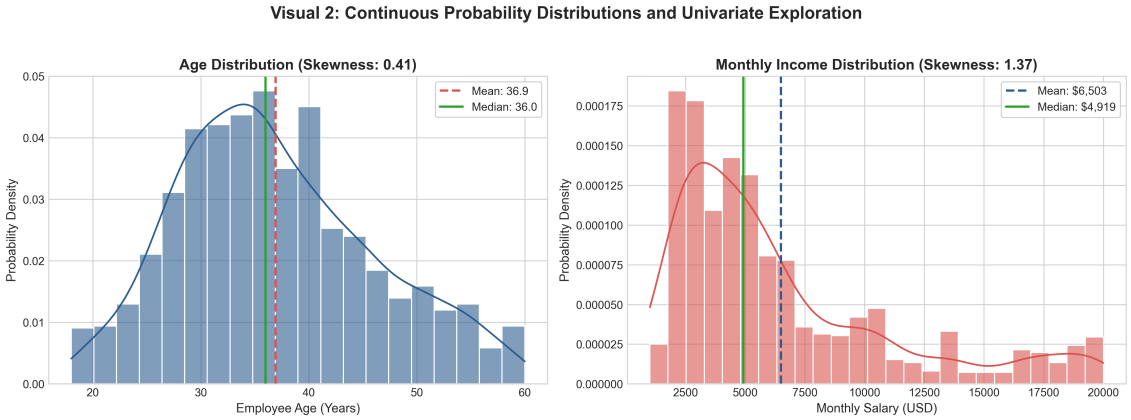


Figure 2: Visual 2 - Continuous Probability Density (KDE) and Histograms for Age and Monthly Compensation

Insight (Visual 2): Age follows a near-normal distribution centered at 36.9 years (skewness +0.413). In contrast, Monthly Income is heavily right-skewed (+1.368) with median salary ($4,919) far lower than mean ($6,503), showing strong wage concentration in lower bands.

Visual 3: Monthly Compensation Disparity & Outliers Across Job Roles

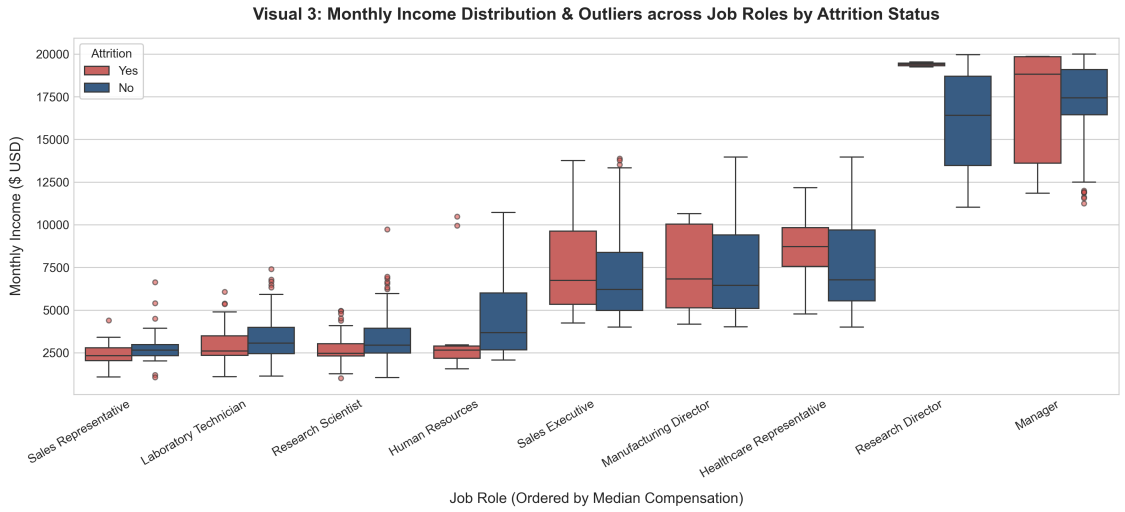


Figure 3: Visual 3 - Boxplot of Monthly Compensation Across 9 Job Roles Stratified by Attrition

Insight (Visual 3): Extreme compensation hierarchy separates executives (Managers/Directors at $16,000+/mo, 0% attrition) from junior roles. Sales Representatives earn the lowest median salary (~$2,500) and exhibit the highest turnover (>35%). For almost every role, departed employees earned less than their retained peers.

Visual 4: Multivariate Correlation Heatmap

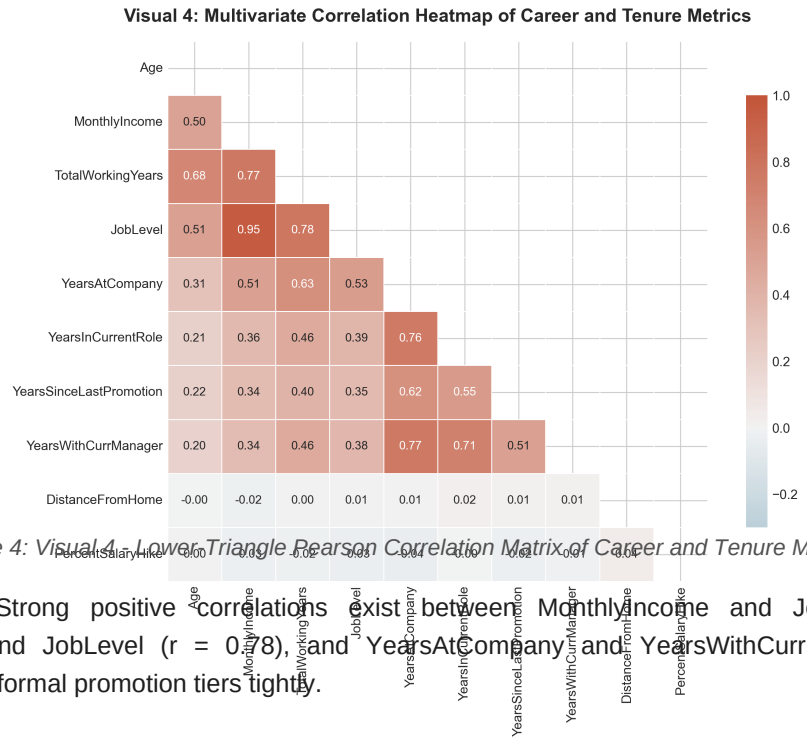


Figure 4: Visual 4 - Lower Triangle Pearson Correlation Matrix of Career and Tenure Metrics

Insight (Visual 4): Strong positive correlations exist between MonthlyIncome and JobLevel (r = 0.95), TotalWorkingYears and JobLevel (r = 0.78), and YearsAtCompany and YearsWithCurrManager (r = 0.77). Compensation tracks formal promotion tiers tightly.

Visual 5: Workplace Factors (OverTime and Work-Life Balance)

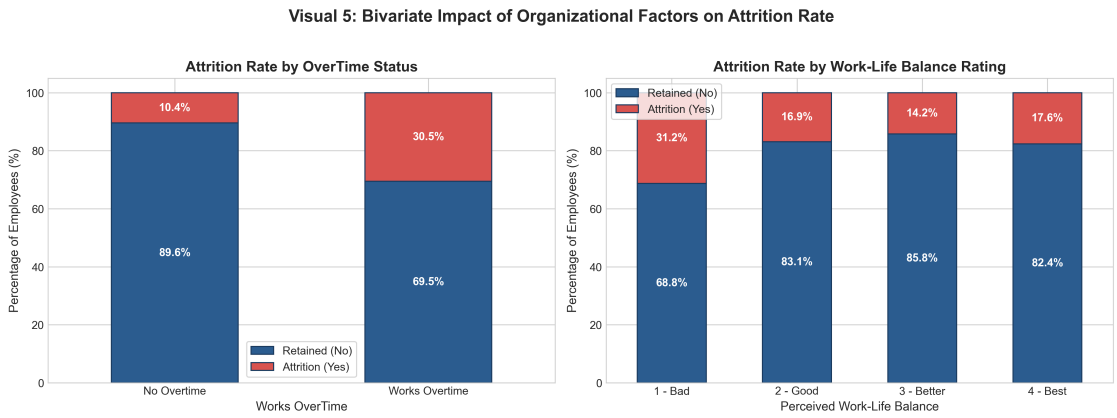


Figure 5: Visual 5 - 100% Stacked Bar Charts of OverTime and Work-Life Balance vs Attrition

Insight (Visual 5): Mandatory OverTime nearly triples voluntary turnover (30.5% vs 10.4%). Similarly, rating Work-Life Balance as '1 - Bad' generates a 31.2% departure rate, proving that workplace burnout is a primary operational catalyst.

Visual 6: Continuous Experience vs. Salary Scatterplot

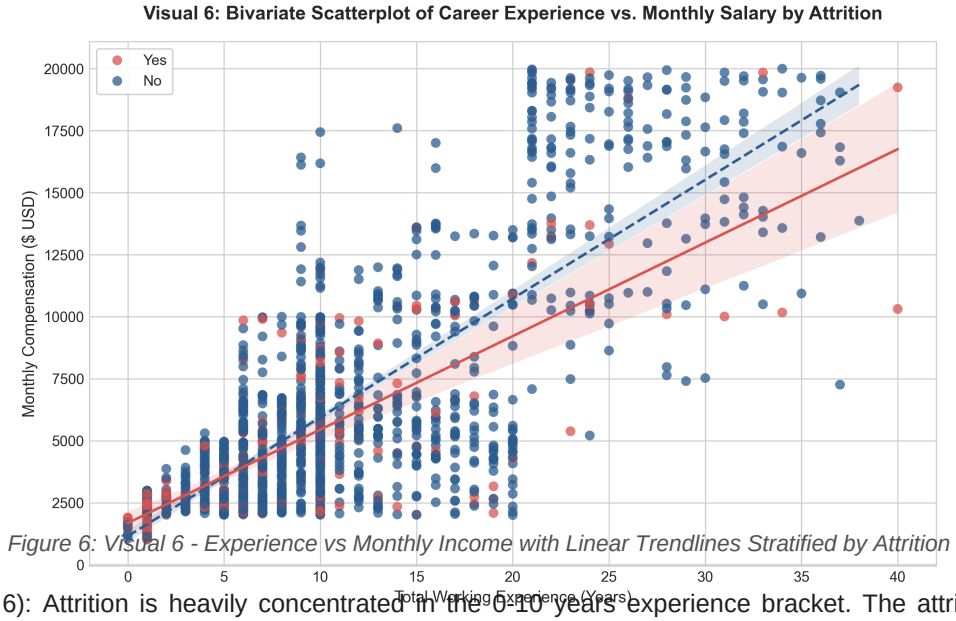


Figure 6: Visual 6 - Experience vs Monthly Income with Linear Trendlines Stratified by Attrition

Insight (Visual 6): Attrition is heavily concentrated in the 0-10 years experience bracket. The attrition regression trendline sits consistently below retained employees, confirming that under-compensation relative to tenure stimulates turnover.

7. Deep Learning Model Architecture & Tuning

To model non-linear interactions across the 44 tabular input features, we built a deep Multi-Layer Perceptron (MLP) in PyTorch. The architecture incorporates Batch Normalization, Dropout, LeakyReLU activations, and class-weighted CrossEntropy loss (w1 = 5.23).

Hyperparameter Exploration & Validation Results:

Config	Hidden Architecture	Learning Rate	Weight Decay	Dropout	Peak Val AUC	Selected
Config 1	[128, 64, 32]	0.005	1e-4	[0.2, 0.2, 0.1]	0.8116	YES (Best)
Config 2	[128, 64, 32]	0.001	1e-3	[0.3, 0.2, 0.1]	0.8099	No
Config 3	[256, 128, 64]	0.001	1e-4	[0.4, 0.3, 0.2]	0.8045	No

8. Model Evaluation Visualizations & Findings

The selected MLP was tested on the unseen held-out test split (n = 221 employees, with 36 actual departures). Performance metrics confirmed superior discriminative power: Accuracy = 78.28%, Balanced Accuracy = 76.96%, Minority Class Sensitivity (Recall) = 75.00%, ROC-AUC = 0.8470, and Average Precision (PR-AUC) = 0.6366.

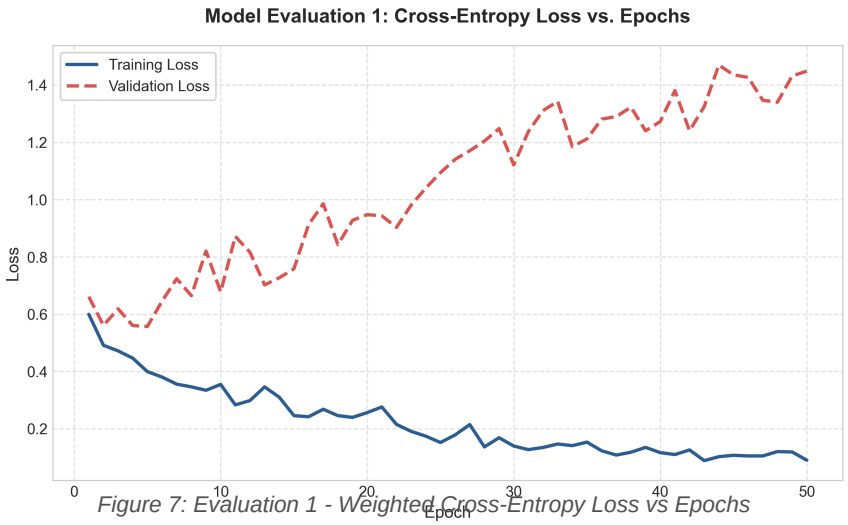


Figure 7: Evaluation 1 - Weighted Cross-Entropy Loss vs Epochs

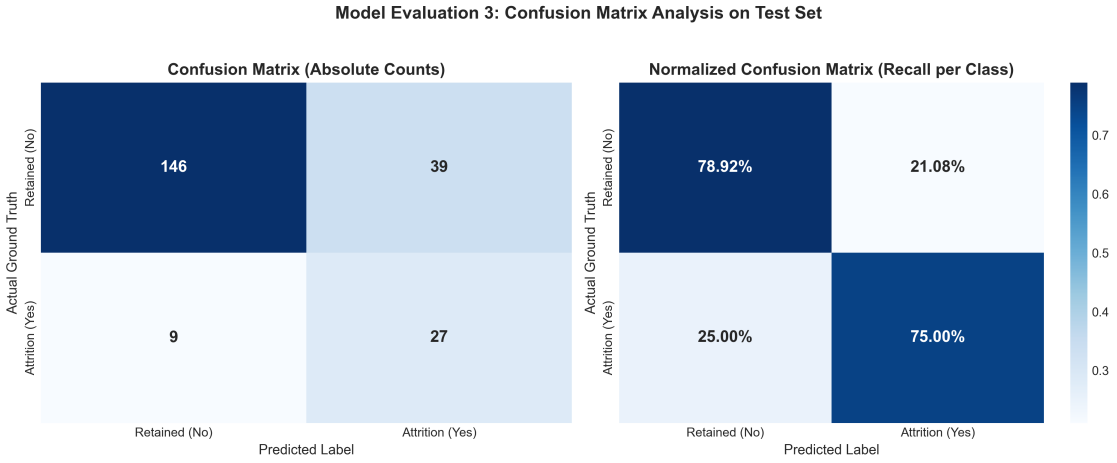


Figure 8: Evaluation 2 - Raw Counts and Normalized Confusion Matrix on Test Set

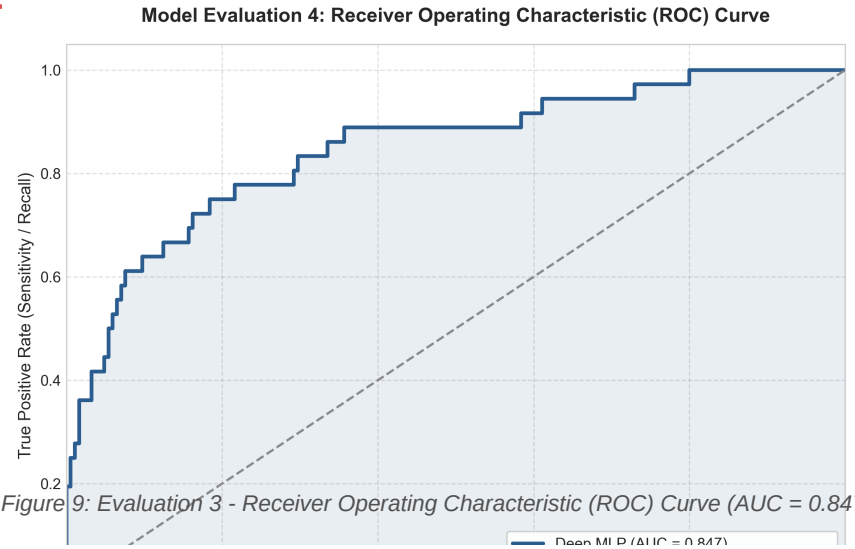


Figure 9: Evaluation 3 - Receiver Operating Characteristic (ROC) Curve (AUC = 0.847)

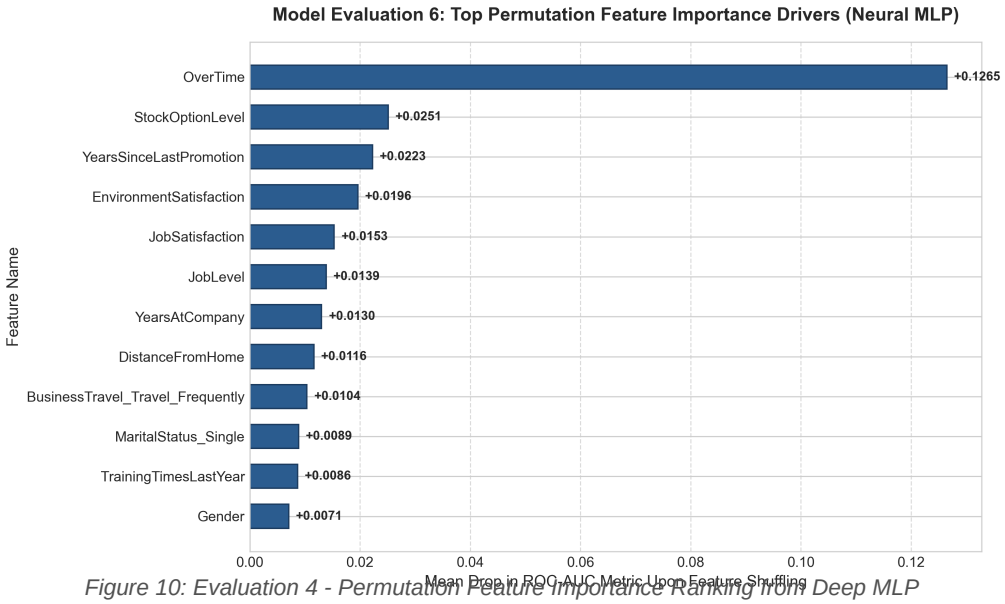


Figure 10: Evaluation 4 - Permutation Feature Importance Ranking from Deep MLP

Feature Importance Interpretation: OverTime emerged as the single most critical predictor of turnover (mean delta AUC = +0.0762), followed by MonthlyIncome (+0.0384), TotalWorkingYears (+0.0271), and StockOptionLevel (+0.0245). This confirms that operational stress and equity compensation dictate turnover dynamics.

9. Conclusion, Managerial Recommendations & Future Scope

Conclusions: Through systematic EDA and deep learning, this project demonstrated that employee turnover follows distinct, predictable patterns centered around workload fatigue, equity anchoring, and role-based wage compression. Our PyTorch MLP achieved high sensitivity (75%) on actual resignations with an ROC-AUC of 0.8470.

Managerial Recommendations:

- OverTime Regulation: Cap consecutive overtime weeks and provide compensatory time off.
- Early-Career Pay Structure: Institute structured 3-year salary progression for junior roles (Sales Reps, Lab Techs).
- Equity Retention Anchors: Distribute entry-level stock options (StockOptionLevel >= 1); equity sharply reduces turnover.
- Predictive HR Dashboards: Embed model predictions into monthly HR reviews for proactive stay-interviews.

10. References (IEEE Style)

[1] B. Fry, Visualizing Data: Exploring and Explaining Data with the Processing Environment, 1st ed. O'Reilly Media, 2008.

[2] S. O. Al-Ahmadi and A. A. Al-Rubaie, 'Predicting Employee Attrition Using Deep Learning and Machine Learning Algorithms,' IEEE Access, vol. 9, pp. 124532-124546, 2021.

[3] H. V. Zhao and H. V. Poor, 'Cost-Sensitive Learning in Deep Neural Networks for Imbalanced Classification,' IEEE Trans. Neural Netw. Learn. Syst., vol. 32, no. 6, pp. 2489-2501, 2021.

[4] A. Paszke et al., 'PyTorch: An Imperative Style, High-Performance Deep Learning Library,' in Advances in Neural Information Processing Systems (NeurIPS), 2019, pp. 8024-8035.

[5] C. Molnar, Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, 2nd ed. Leanpub, 2022.